water in ports and harbors.

Polluting the oceans

Microplastics—tiny fragments of plastic less than ¼ inch (5 mm) across—are even harder to clean up than other plastics. Comprising 90 percent of the plastics in the oceans, they surge through currents like a murky soup. The problem was first identified in 1997 by the American oceanographer Captain Charles Moore, who highlighted it in his 2011 book *Plastic Ocean*. Sailing home from a yachting competition, Moore came across a vast patch of plastic debris in the Pacific Ocean. Now known to have a surface area three times the size of continental France, the Great Pacific Garbage Patch (GPGP) comprises 80,000 metric tons of plastic, amassed by the swirling current known as the North Pacific Gyre.

The GPGP is one of several oceanic garbage patches—there are others in the Atlantic and Indian Ocean as well as in smaller bodies of water such as the North Sea. Plastic microbeads, introduced by cosmetic companies in the 1990s, add to the problem. Used in personal care products such as soaps, facial scrubs, and toothpastes, the beads travel from wastewater systems into rivers and oceans, where they are consumed by fish and other animals, with the same damaging effects as microplastics (see Effects on wildlife).